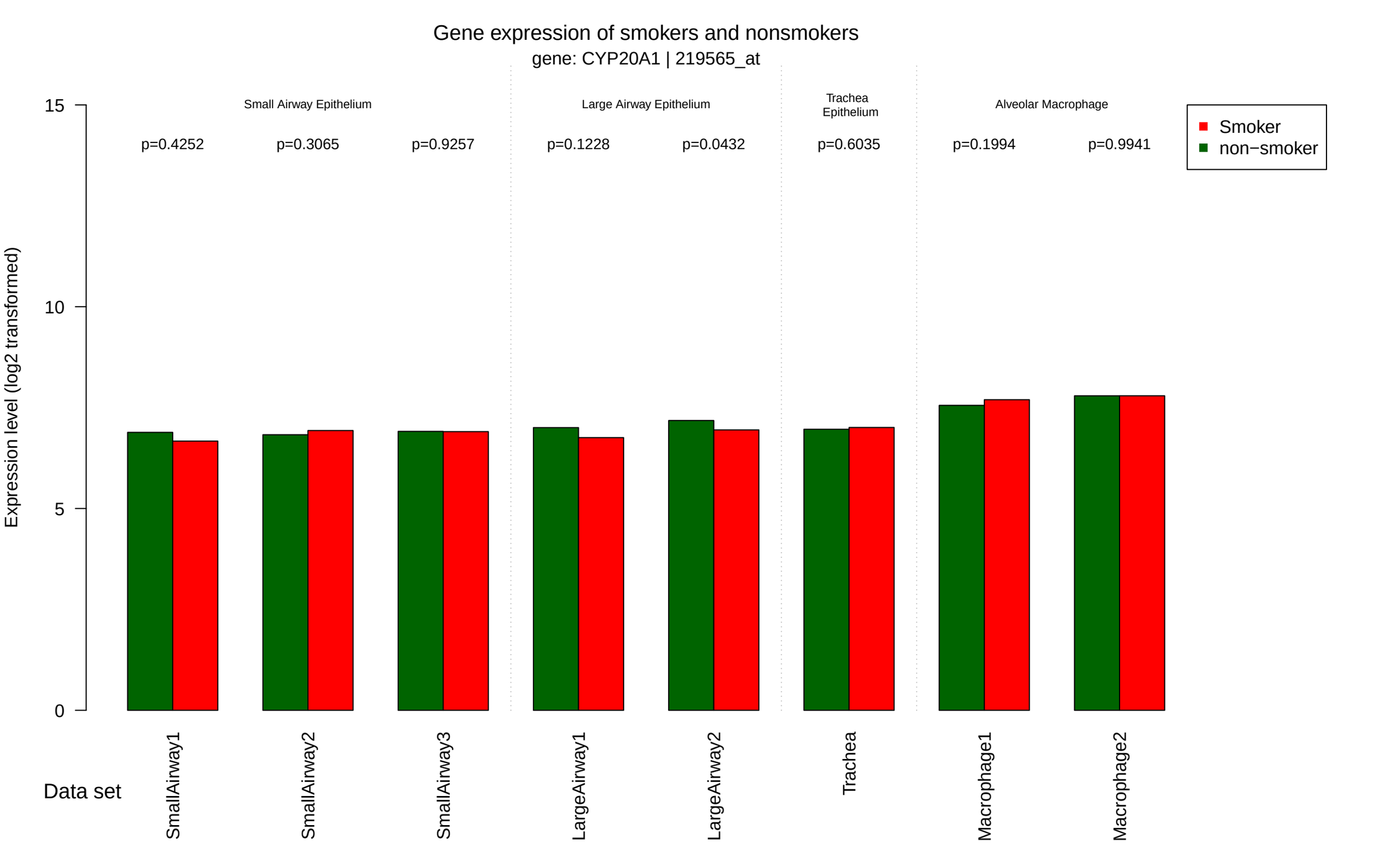
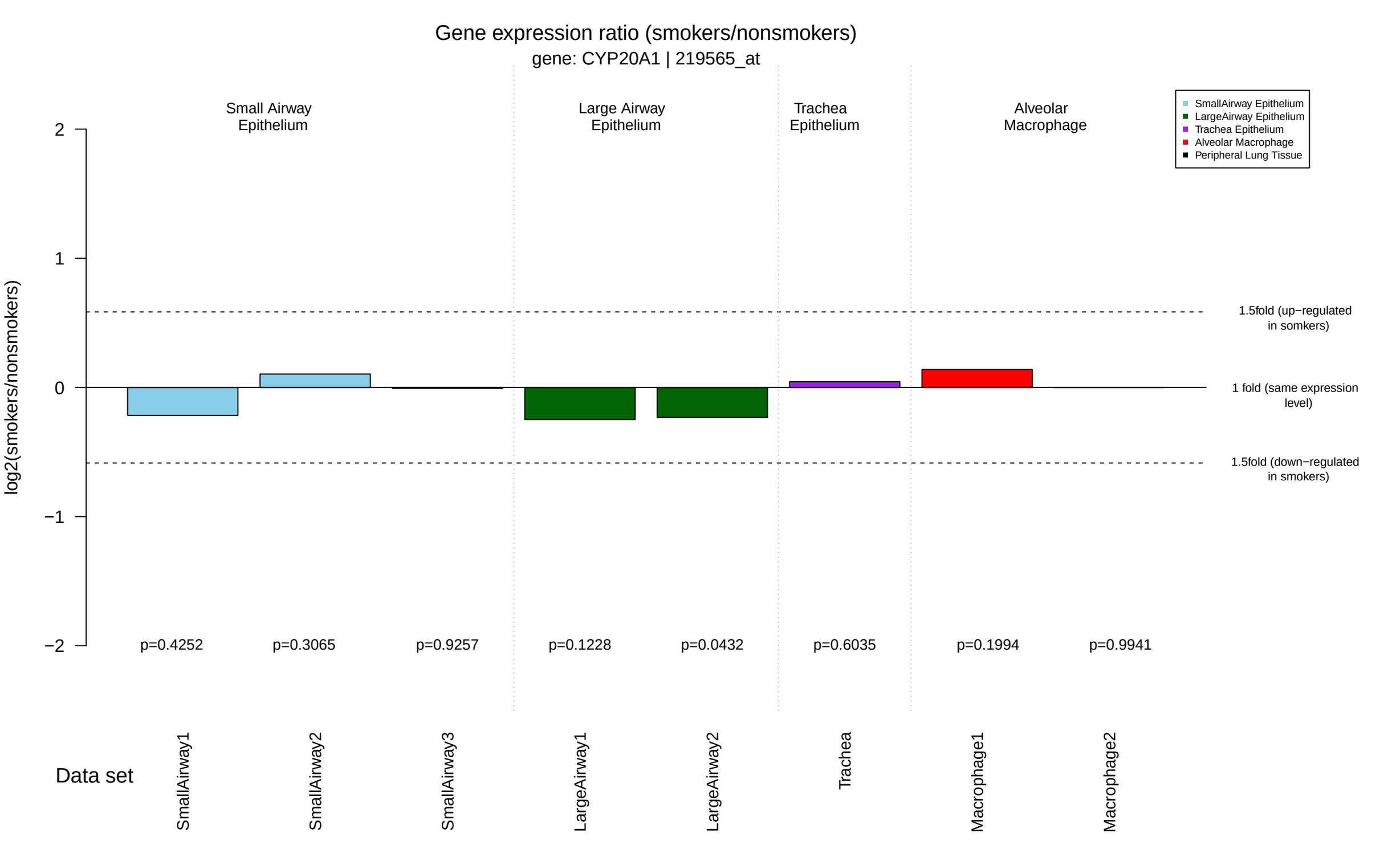
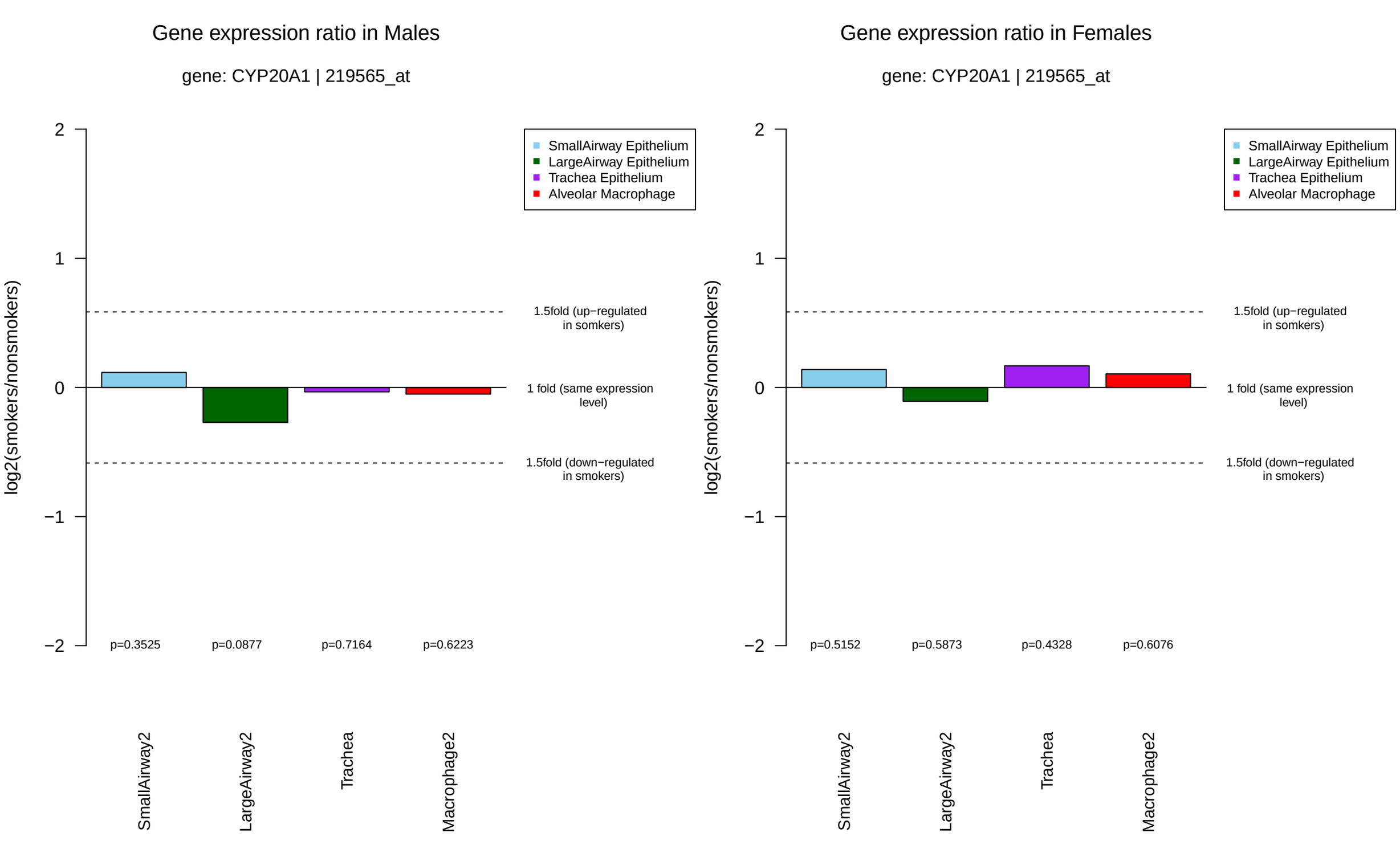
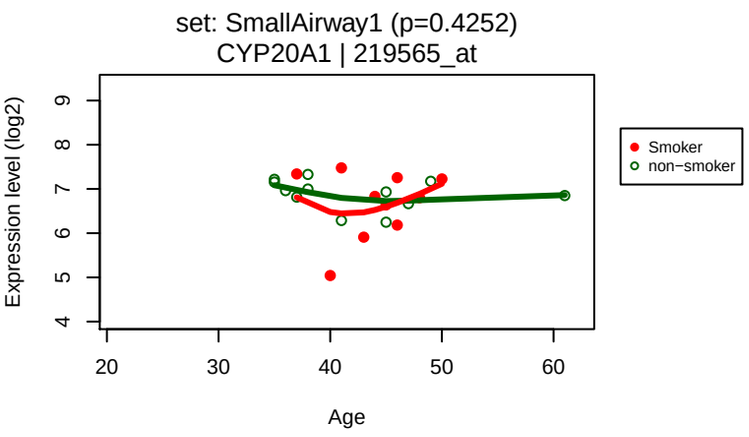
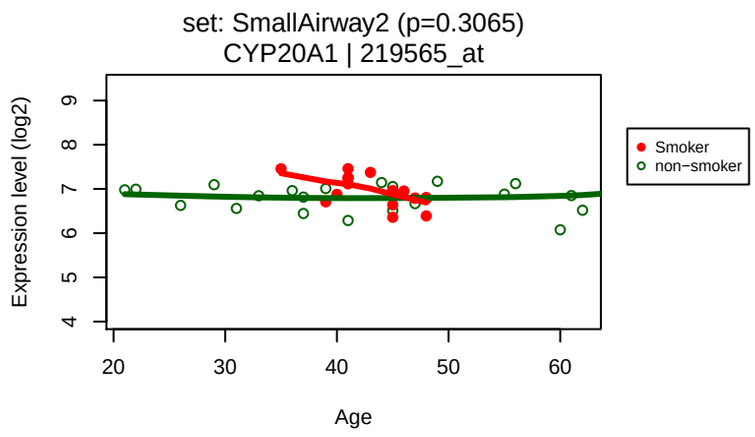
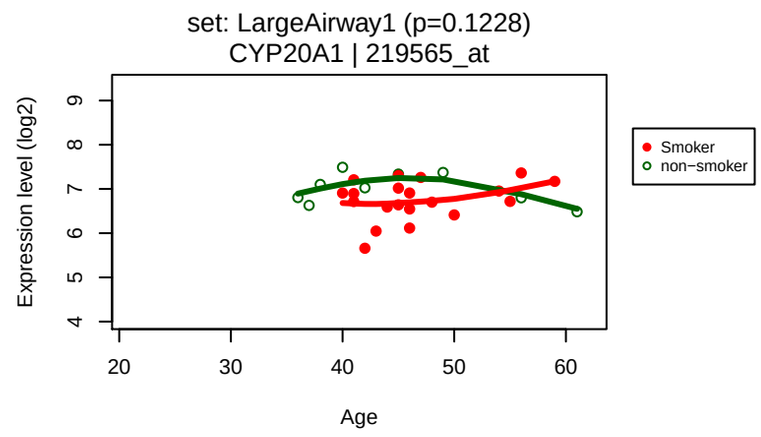

Gene expression of smokers and nonsmokers
gene: CYP20A1 | 219565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4252
p=0.3065
p=0.9257
p=0.1228
p=0.0432
p=0.6035
p=0.1994
p=0.9941
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

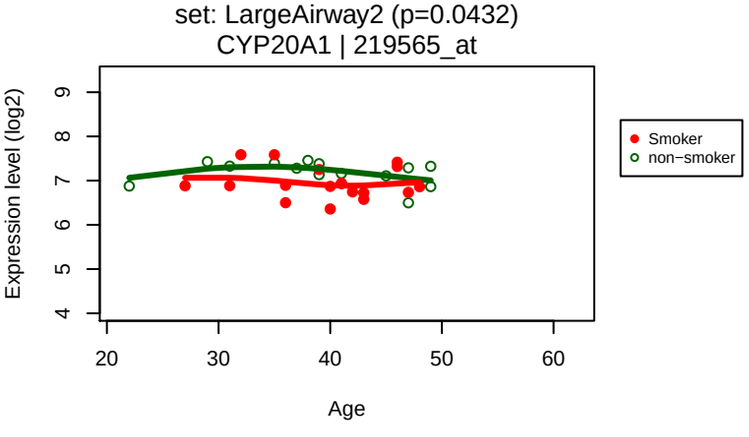

Gene expression ratio (smokers/nonsmokers)
gene: CYP20A1 | 219565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4252
p=0.3065
p=0.9257
p=0.1228
p=0.0432
p=0.6035
p=0.1994
p=0.9941
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

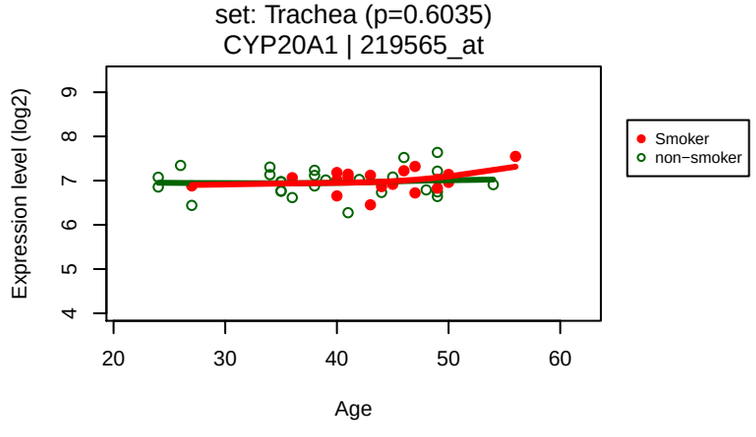

Gene expression ratio in Males
gene: CYP20A1 | 219565_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3525
p=0.0877
p=0.7164
p=0.6223
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP20A1 | 219565_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5152
p=0.5873
p=0.4328
p=0.6076
SmallAirway2
LargeAirway2
Trachea
Macrophage2

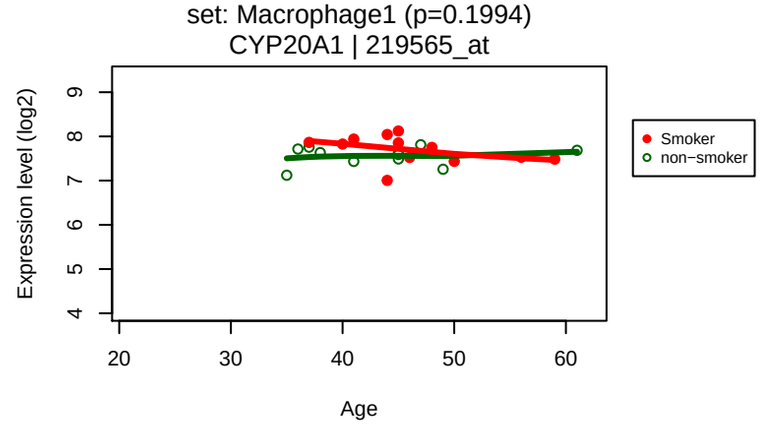

set: SmallAirway1 (p=0.4252)
CYP20A1 | 219565_at
Expression level (log2)
Age
Smoker
non−smoker

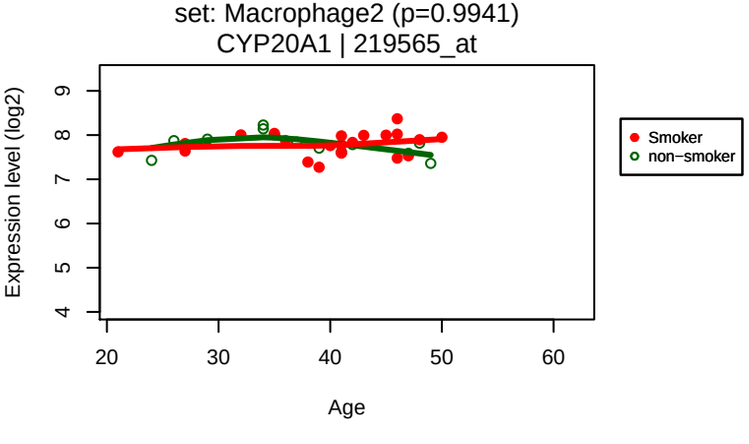

set: SmallAirway2 (p=0.3065)
CYP20A1 | 219565_at
Expression level (log2)
Age
Smoker
non−smoker

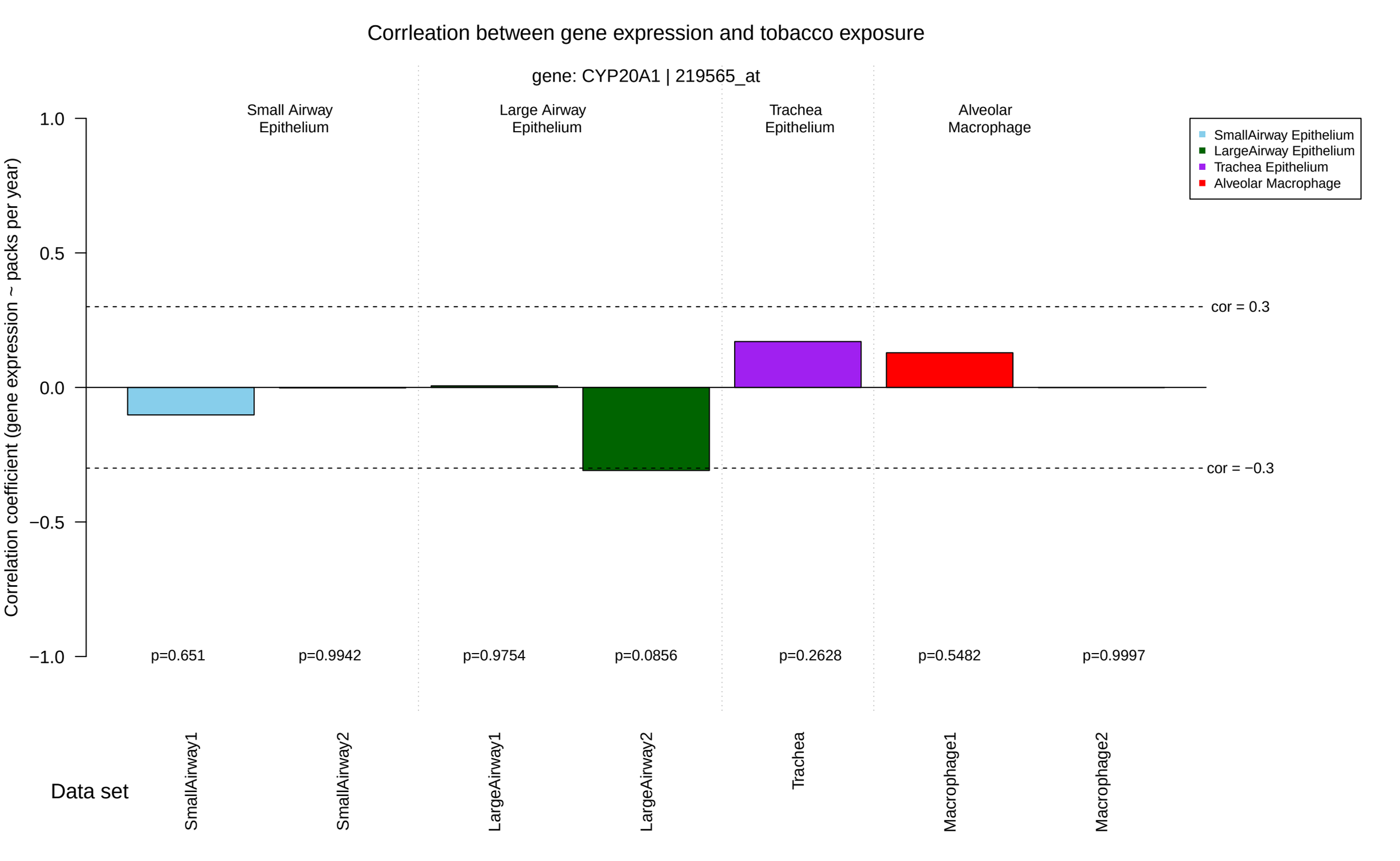

set: LargeAirway1 (p=0.1228)
CYP20A1 | 219565_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0432)
CYP20A1 | 219565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6035)
CYP20A1 | 219565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1994)
CYP20A1 | 219565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9941)
CYP20A1 | 219565_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP20A1 | 219565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.651
p=0.9942
p=0.9754
p=0.0856
p=0.2628
p=0.5482
p=0.9997
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2